

Year 10 Science Physical Sciences Worksheet - Speed and Velocity

SPEED

The speed of a body is the rate of change of distance by the body.

The average speed of a body over a distance travelled in a time interval can be determined by:

$$\text{Speed (av)} = \frac{d}{t}$$

where d = distance (measured from zero)
 t = time interval (measured from zero)

Speed is a scalar quantity as it involves no specific direction. Distance is the total length of the path covered by the body.

If to accurately describe motion direction is required, then **VELOCITY** is used in place of speed.

VELOCITY

The velocity of a body is the rate of change of displacement of the body.

The average velocity of a body over a displacement in a measured time interval can be determined by:

$$v_{av} = \frac{s}{t}$$

v_{av} = average velocity
 s = displacement (measured from zero)
 t = time interval (measured from zero)

Velocity is a vector quantity. *The direction of the velocity is that of the displacement (unless otherwise stated).*

UNITS OF VELOCITY AND SPEED

Velocity and speed are both determined by dividing units of length (m) by units of time (s).

$$\text{Units: } \frac{\text{metres}}{\text{seconds}} = \frac{\text{m}}{\text{s}} = \text{m s}^{-1} \text{ (or m/s)}$$

DETERMINATION OF DISPLACEMENT

The equation for average velocity can be rearranged to determine the displacement.

$$v_{av} = \frac{s}{t}$$

$$\begin{array}{ll} \text{rearranging gives:} & s = v_{av} \times t \\ \text{similarly} & d = \text{speed (av)} \times t \end{array}$$

TYPE EXAMPLE 7: Calculate the average velocity of a vehicle which travels 3 km east in 5 minutes.

$$\begin{array}{l} v_{av} = ? \\ s = 3 \text{ km} \\ \quad = (3 \times 1\,000) \text{ m} \\ t = 5 \text{ minutes} \\ \quad = (5 \times 60) \text{ s} \end{array}$$

$$\begin{array}{l} v_{av} = \frac{s}{t} \\ \quad = \frac{3 \times 1\,000}{5 \times 60} \\ \quad = 10 \text{ m s}^{-1} \end{array}$$

i.e. the average velocity is 10 m s^{-1} east.

TYPE EXAMPLE 8: Determine the displacement of a body which moves with a uniform velocity of 2.4 m s^{-1} for 50 s .

$$v_{av} = 2.4 \text{ m s}^{-1}$$

$$s = ?$$

$$t = 50 \text{ s}$$

$$v_{av} = \frac{s}{t}$$

$$\begin{aligned}\therefore s &= v_{av} \times t \\ &= 2.4 \times 50 \\ &= 120 \text{ m}\end{aligned}$$

i.e. the displacement is 120 m in the direction of the velocity.

SET 6: VELOCITY I

1. What is the average speed of a sprinter who runs 100 m in 12.5 s ?
2. A car takes 80 s to complete one lap of a 2 km circuit. What is its average speed (in m s^{-1})?
3. A body travels 36 m in 4 s . What is its average speed?
4. How far will a bird fly in 25 s travelling with an average speed of 6 m s^{-1} ?
5. A person swims from one end of a pool to the other at an average velocity of 1.5 m s^{-1} in 20 s . What is the length of the pool?
6. Determine the distance travelled by a plane in 1 hour if it flies due south with a uniform velocity of 90 m s^{-1} .
7. How long will it take for a body moving with a uniform velocity of 7 m s^{-1} to cover 98 m ?
8. A cricket ball is thrown a distance of 90 m from the boundary to the wicket at an average horizontal speed of 30 m s^{-1} . How long will it take for the ball to reach the wicket?
9. Determine the time taken for a horse running with an average speed of 14 m s^{-1} to complete one circuit of a 840 m track.
10. How long will it take to complete a trip from Perth to Kalgoorlie, a distance of 595 km , travelling at an average speed of 70 km h^{-1} ?
11. A train moving with an average speed of 80 km h^{-1} takes 0.25 hours to travel from one railway siding to the next. What is the distance between the two sidings?
12. Determine the average speed (km h^{-1}) of a plane which flies a distance of 720 km in 1.5 hours ?

SET 6

1. 8 m s^{-1}

6. $324\,000 \text{ m}$

11. 20 km

2. 25 m s^{-1}

7. 14 s

12. 480 km h^{-1}

3. 9 m s^{-1}

8. 3 s

4. 150 m

9. 60 s

5. 30 m

10. 8.5 h

SET 7: VELOCITY II

1. An athlete, commencing at a point due east, completes half a lap of a 440 m track in 25 s. Determine the athlete's:
 - (a) average speed.
 - (b) average velocity (in westerly direction).
 - (c) average velocity if the athlete completes one full lap at the same pace.
2. A car travels with an average speed of 20 m s^{-1} due north for 3 km then due west for another 4 km calculate:
 - (a) the total time taken for the journey.
 - (b) the displacement of the car.
 - (c) the average velocity of the car.
3. A person walks for 10 s at 1 m s^{-1} then for another 10 s in the same direction at 2 m s^{-1} . Find the:
 - (a) total displacement.
 - (b) average velocity.
4. A body travels 200 m at a uniform velocity of 4 m s^{-1} and a further 150 m in the same direction at a uniform velocity of 7.5 m s^{-1} . Determine the:
 - (a) total time taken.
 - (b) average velocity for the whole journey.
5. A cyclist travels at 6 m s^{-1} S for 10 minutes and then returns 600 m along the same path. What is the final displacement of the cyclist?
6. In a 50 m swimming race the winner completed the race in 32 seconds. By what margin (length) did the winner defeat the second placegetter who swam with an average velocity of 1.5 m s^{-1} ?
7. A cyclist moving uniformly at 5 m s^{-1} overtakes a pedestrian walking in the same direction in 6 s. If the pedestrian initially was 18 m ahead of the cyclist, find the pedestrian's average velocity.
8. Determine the time saved by travelling 8 km at 40 m s^{-1} rather than at 32 m s^{-1} .
9. How far will a car moving at 20 m s^{-1} travel in the same time that a car moving at 25 m s^{-1} travels 1 km?
10. A car travelling at 15 m s^{-1} is 10 m behind another vehicle moving in the same direction. If it takes 10 s to overtake this vehicle, what is the speed of the slower vehicle?

SET 7

1. (a) 8.8 m s^{-1} (b) 5.6 m s^{-1} W (c) 0 2. (a) 350 s (b) 5 km N 53° W
(c) 14.3 m s^{-1} N 53° W 3. (a) 30 m (b) 1.5 m s^{-1} 4. (a) 70 s (b) 5 m s^{-1}
5. 3 000 m S 6. 2 m 7. 2 m s^{-1} 8. 50 s 9. 800 m 10. 14 m s^{-1}